

Drone Project - First Engineering Worksheet

This English version is formatted for graduate application review. It preserves the engineering purpose, design decisions, validation evidence, and individual contribution signals of the original course document while presenting the material in concise English.

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Project	Quadcopter Ball-Transfer Drone
Course	Practice of Mechanical Engineering
Document Type	Worksheet
Primary Role	Team lead and concept contributor
Original Source	drone-first-project-worksheet.pdf (6 pages)

Executive Summary

This worksheet supported early requirement definition, concept comparison, and team planning for the ball-transfer drone project.

It is included as process evidence showing that the final system was developed through structured engineering checkpoints rather than a single late build attempt.

Applicant Contribution Signals

- Defined early mission requirements and design questions.
- Compared feasible build directions before detailed fabrication.
- Established a shared language for later subsystem work.

Worksheet Function

Record early assumptions, candidate approaches, expected risks, and next tasks.

Translate a broad course challenge into specific engineering decisions for the team.

Technical Decision Table

Constraint	Decision	Why It Mattered	Skill Demonstrated
Open-ended mission	Outlined requirements and constraints	Made design work concrete	Requirements analysis
Multiple concepts	Compared early paths	Prepared better prototype choices	Concept selection
Team coordination	Recorded next actions	Kept work aligned	Project management

How to Read This Evidence

This document is intended to help a reviewer quickly understand the engineering content of the original report in English. The original PDF remains available in the project archive as the source artifact with its original figures, tables, screenshots, and course formatting.